Reminders

<u>Upcoming due dates</u>
Wed Mar 11 A4
Fri Mar 18 Final project*, video*, team eval survey, post-course survey

How to be wrong IV

Data Science in Practice

Jason G. Fleischer, PhD **Dept. of Cognitive Science** **UC San Diego**

Hi,

We are researchers in the UCSD CSE department conducting a study on computational notebooks. We are looking for students in intro data science classes using Jupyter Notebooks and Python. If you're interested in participating in a **1-1.5 hour, in-person coding study**, please fill out this form. Participants will be compensated **$20/hr**.

If you have any questions, please contact akurdak@ucsd.edu.

Thank you for your interest!

Ayla Kurdak, PhD Student
Michael Coblenz, Assistant Professor

Seeking PLAs for next quarter

- Holding office hours / answering student questions
- Helping write / test new assignments
- Collecting class demos, notes, videos and organizing into an online textbook
- Assisting with project feedback

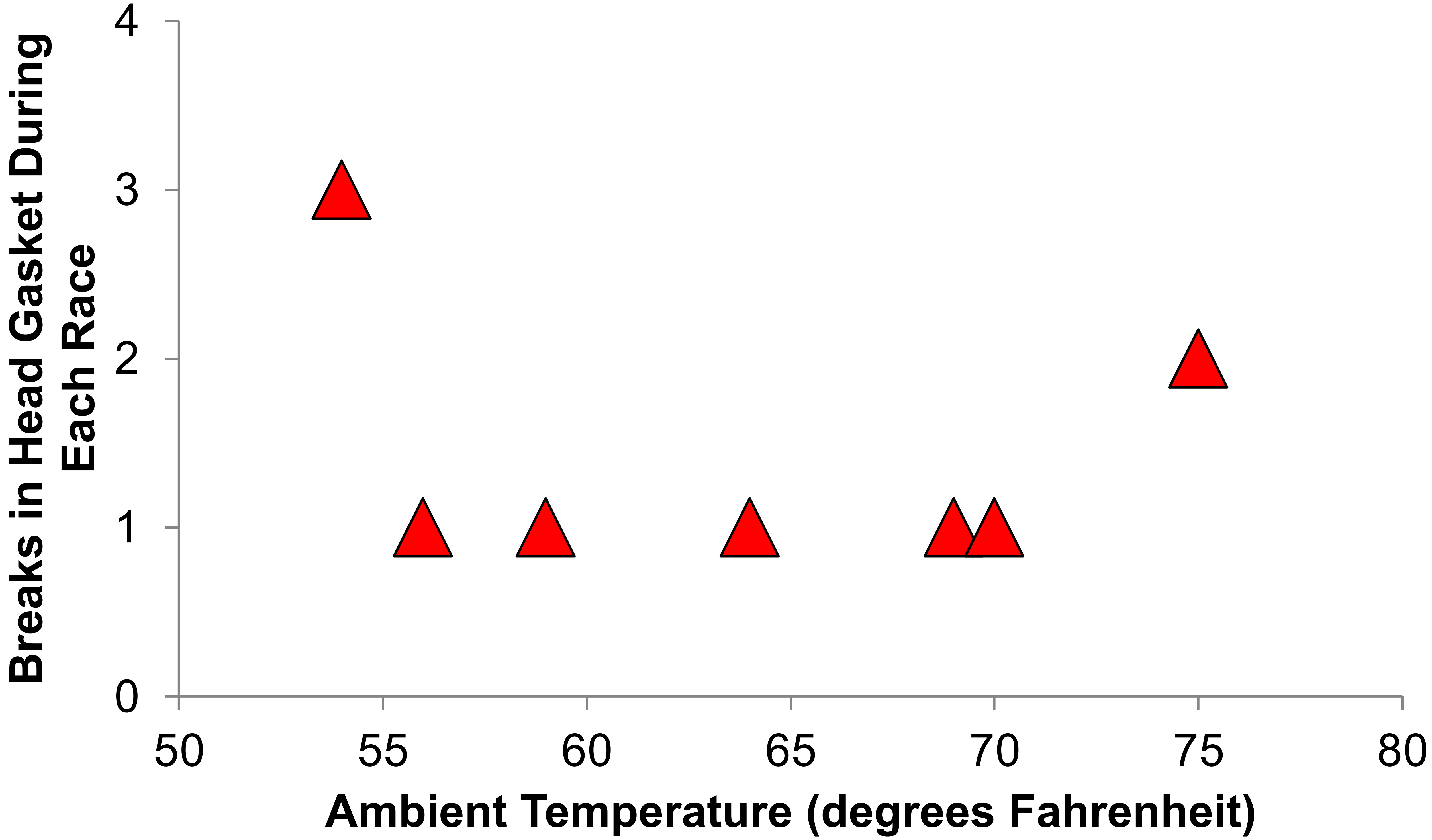
Space Shuttle Challenger

- 73 seconds into the 25th shuttle flight (Jan 28, 1986) suffered a failure of the SRB o-rings caused by low temperature at launch (26 deg F)
- 7 dead: Scobee, Smith, McNair, Resnik, Jarvis, McCauliffe; estimated $3.2 billion loss
- NASA was operating under severe budget pressure. The decision makers involved considered the shuttle program’s success vital to future congressional support.
- This was a media event to drum up more support for the program. Prior delays had pushed the Challenger launch back to the point where it would have to be scrubbed because of other scheduled launches (i.e., there were no second chances).
- Within NASA shuttle launches were widely considered routine and the vehicle safe.
- The record of 24 successful launches was a powerful decision context. Because of these previous successes, engineers were asked to prove a failure was going to occur, not that success was questionable

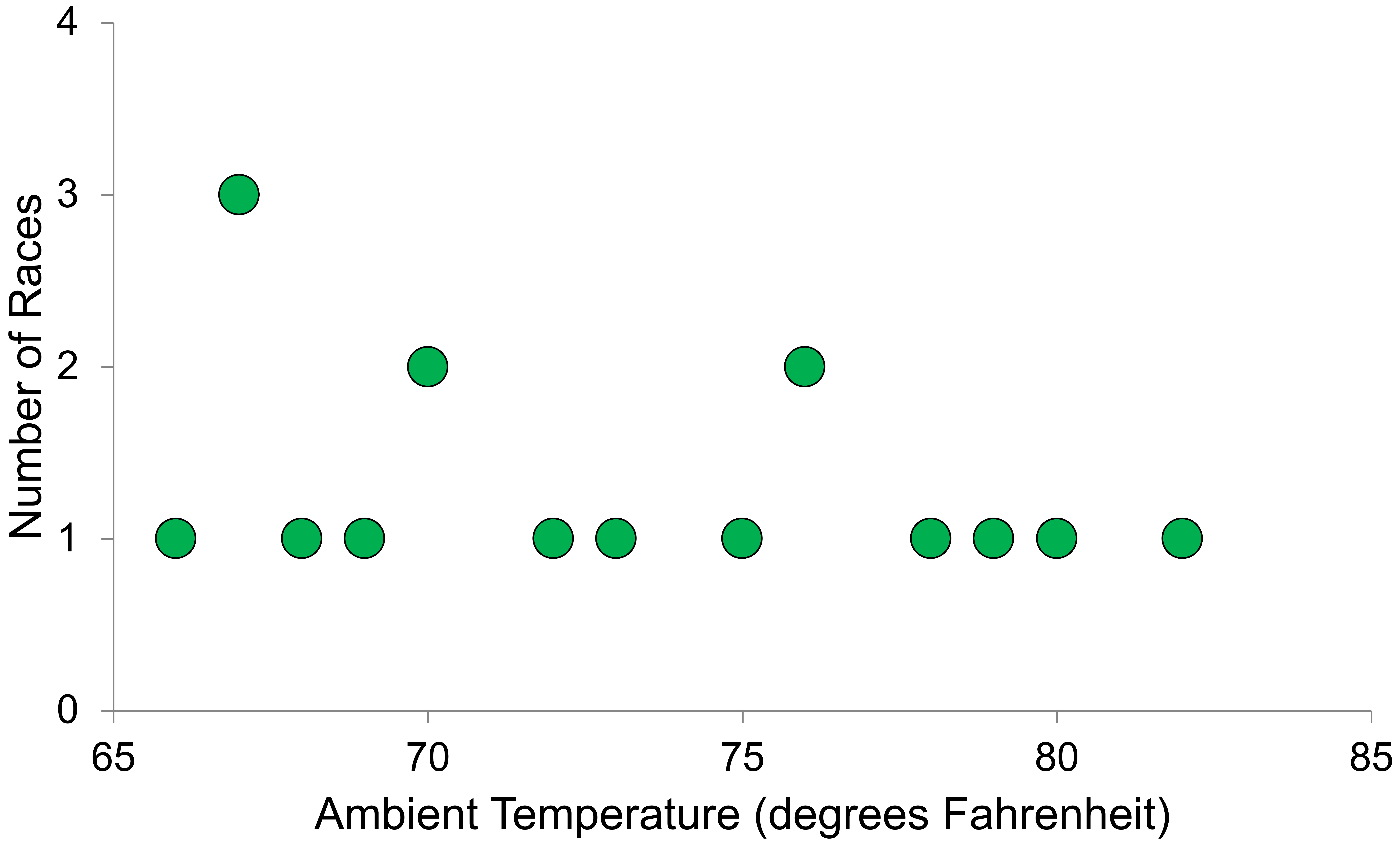
Errors of communication: Teamwork and comm failures

Note from Tom Davis

I got the data on the gasket failures from Antonio. We have run **24 races** this season, with **temperatures at race time ranging from 53 to 82 degrees**. Antonio had a good idea in suggesting we look into this, but as you can see, this is not our problem. I tested the data for a correlation between temperatures and gasket failures, and found no relationship.



Temperature and *no* gasket failure



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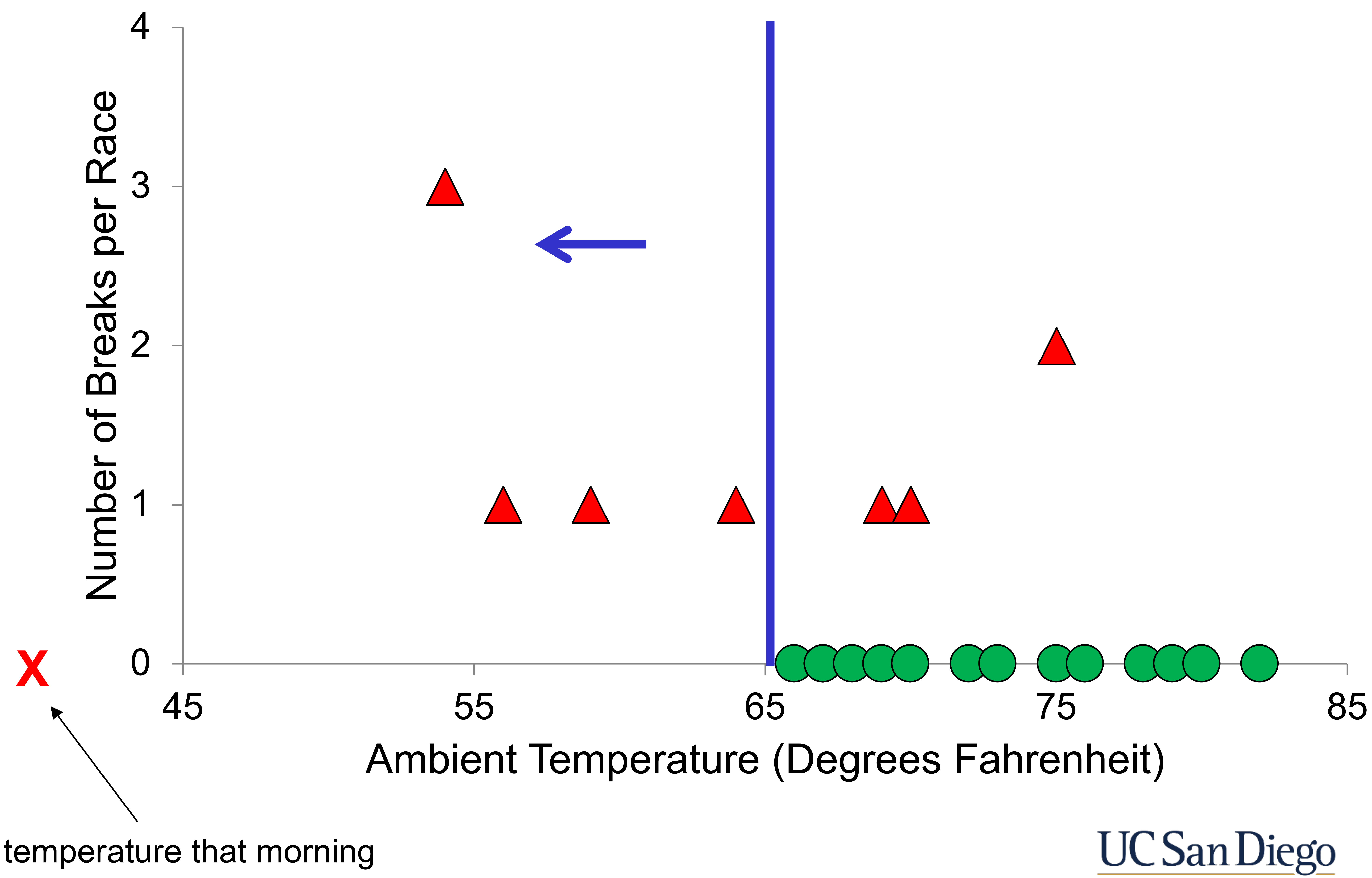
Temperature and all races



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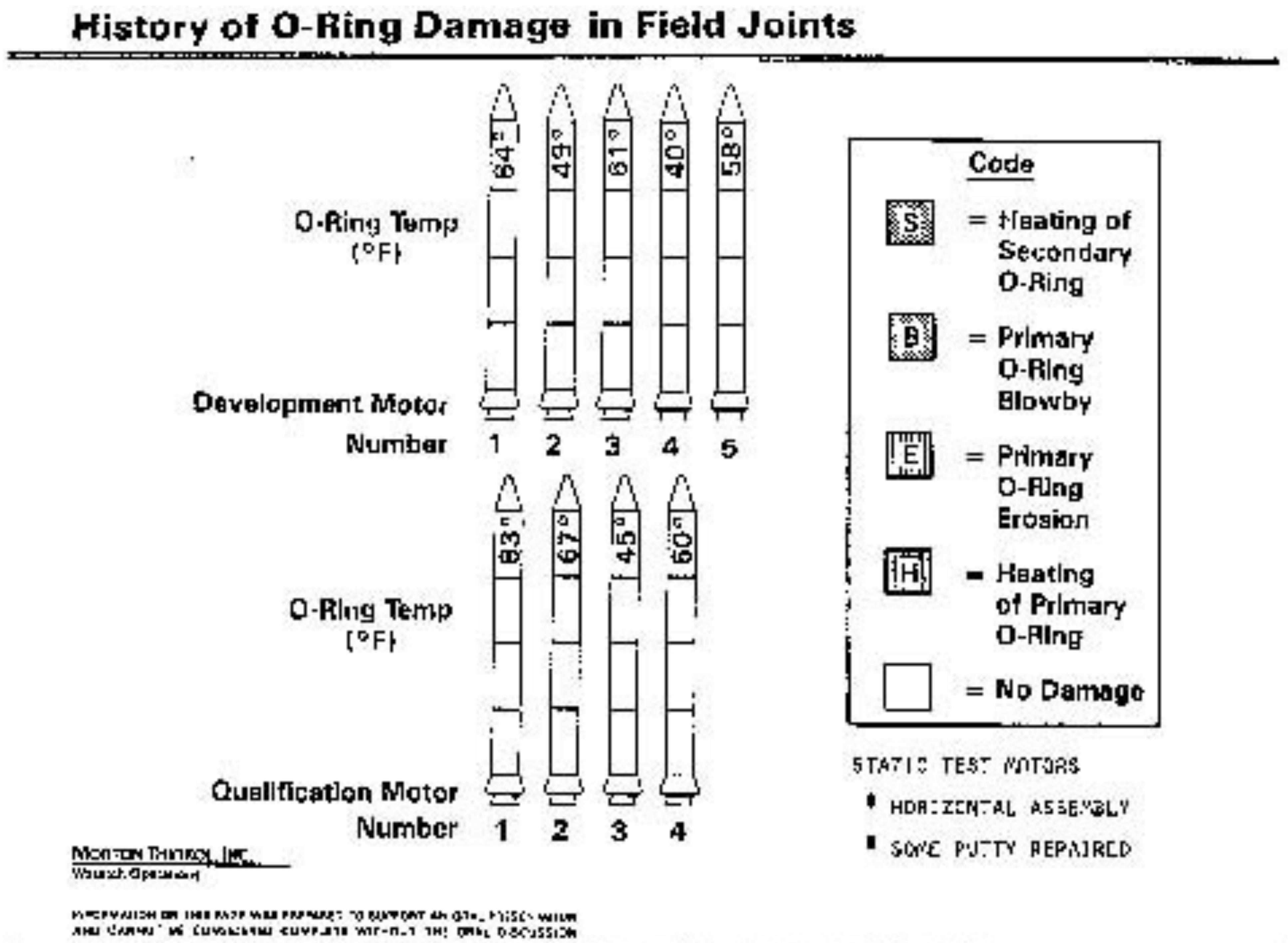
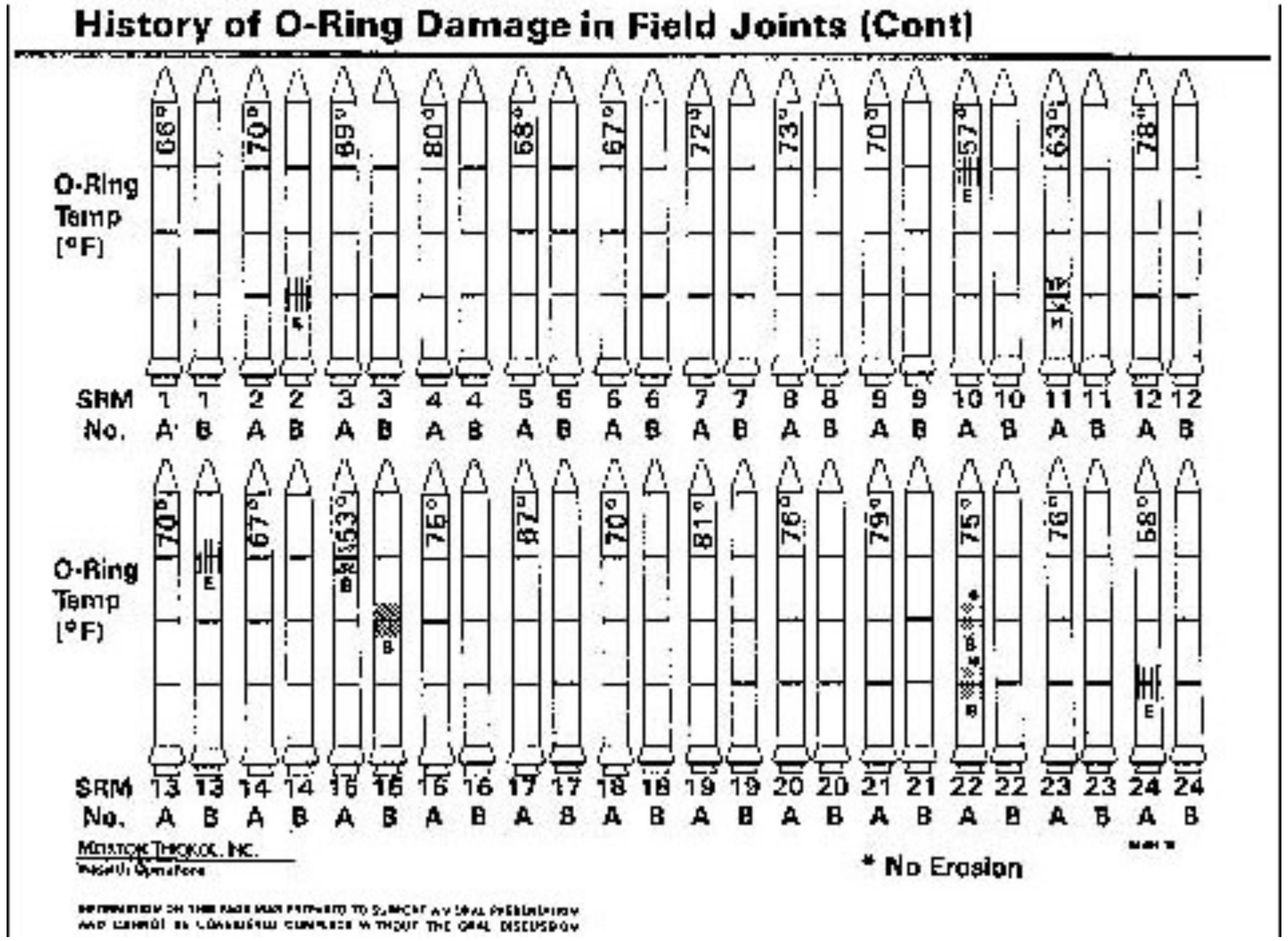
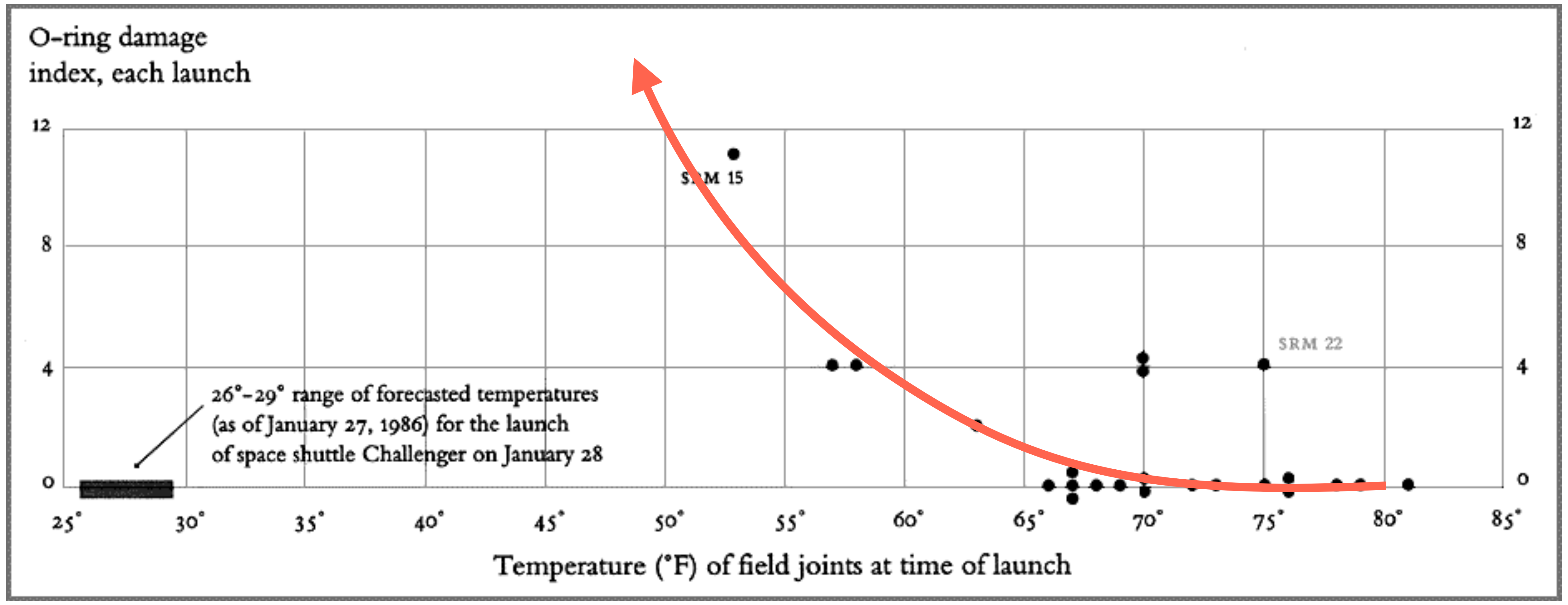
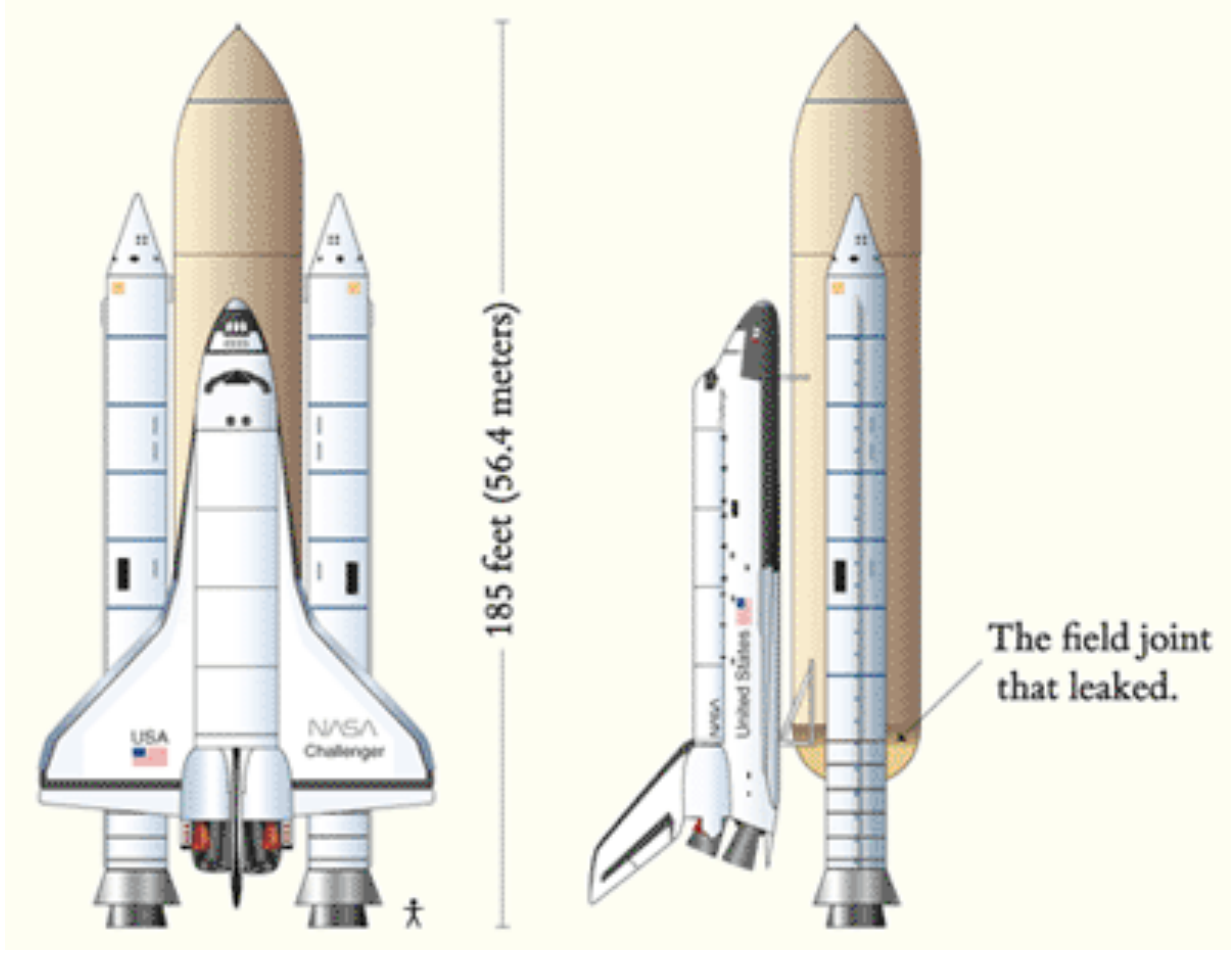
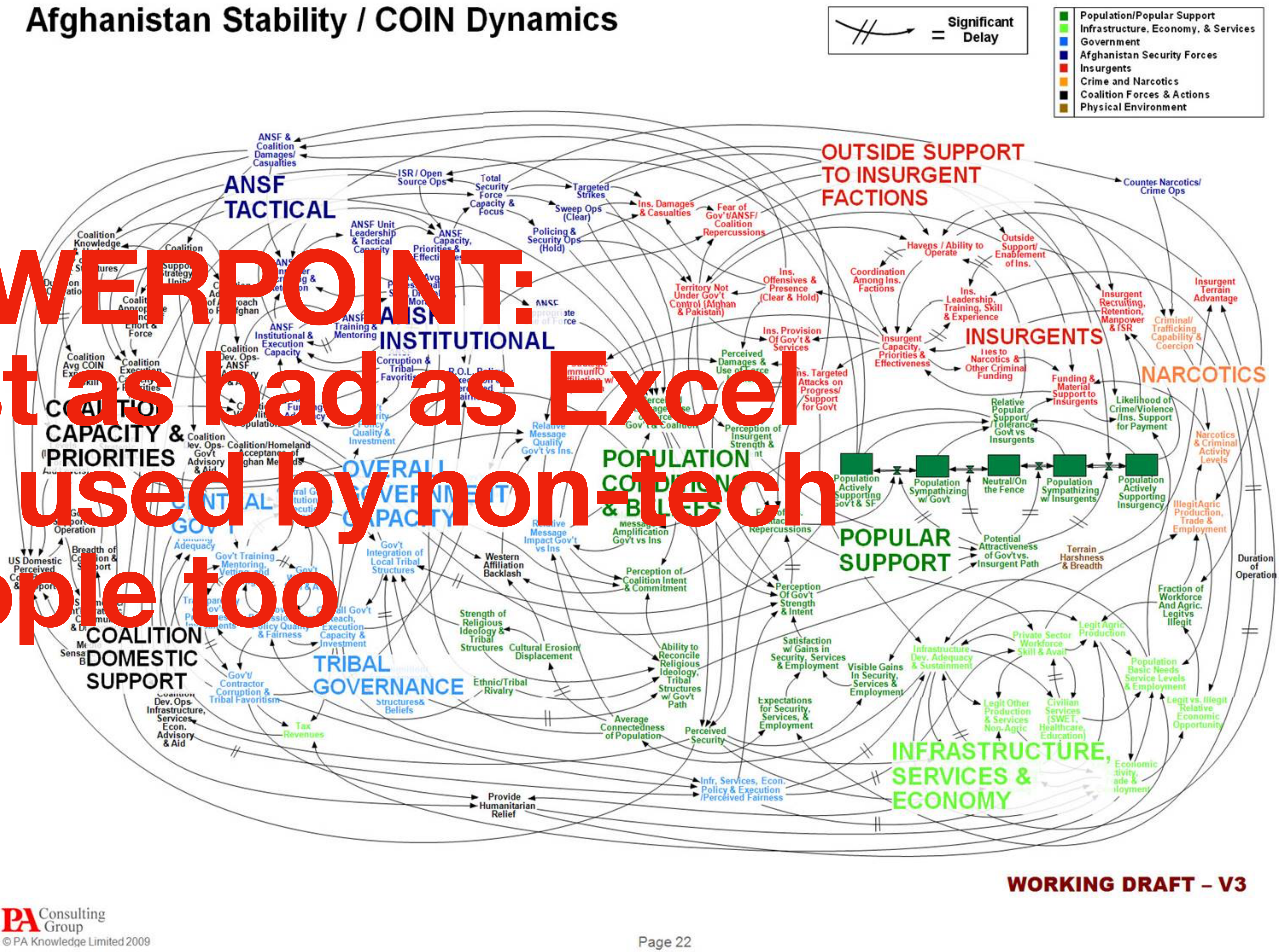






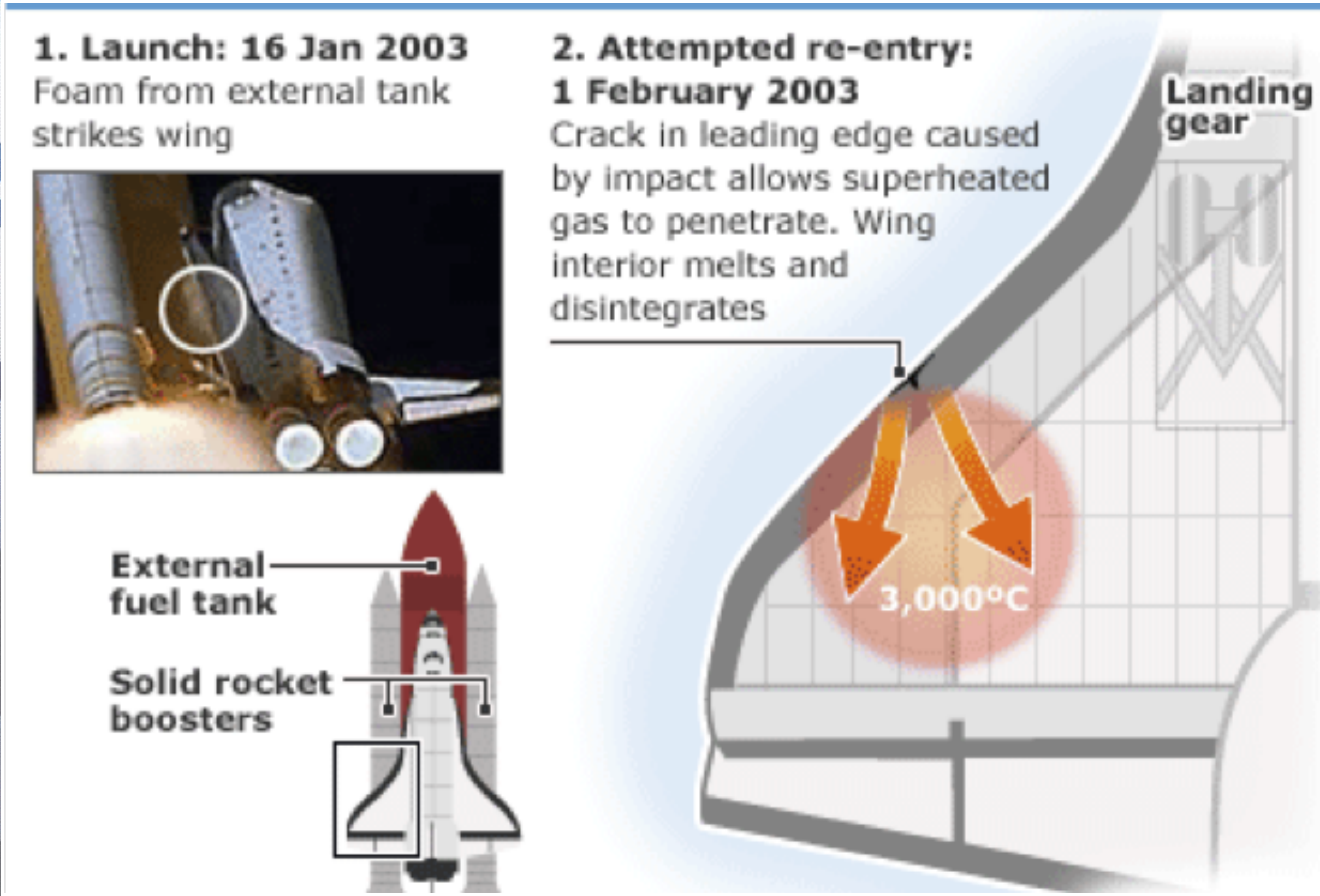
Images and graphs from Edward Tufte





POWERPOINT: Just as bad as Excel but used by non-tech people too


BROWN
CLARK
CHAWLA
ANDERSON
RAMON
HUSBAND
MCCOOL
STS 107


1. Launch: 16 Jan 2003
Foam from external tank strikes wing
External fuel tank
Solid rocket boosters
2. Attempted re-entry: 1 February 2003
Crack in leading edge caused by impact allows superheated gas to penetrate. Wing interior melts and disintegrates
Landing gear
3,000°C

On this one Columbia slide, a PowerPoint festival of bureaucratic hyper-rationalism, 6 different levels of hierarchy are used to display, classify, and arrange 11 phrases:

Level 1		Title of Slide
Level 2	●	Very Big Bullet
Level 3	–	big dash
Level 4	◆	medium-small diamond
Level 5	•	tiny square bullet
Level 6	( )	parentheses ending level 5

The analysis begins with the dreaded Executive Summary, with a conclusion presented as a headline: "Test Data Indicates Conservatism for Tile Penetration." This turns out to be unmerited reassurance. Executives, at least those who don't want to get fooled, had better read far beyond the title.

The "conservatism" concerns the *choice of models* used to predict damage. But why, after 112 flights, are foam-debris models being calibrated during a crisis? How can "conservatism" be inferred from a loose comparison of a spreadsheet model and some thin data? Divergent evidence means divergent evidence, not inferential security. Claims of analytic "conservatism" should be viewed with skepticism by presentation consumers. Such claims are often a rhetorical tactic that substitutes verbal fudge factors for quantitative assessments.

As the bullet points march on, the seemingly reassuring headline fades away. Lower-level bullets at the end of the slide undermine the executive summary. This third-level point notes that "Flight condition [that is, the debris hit on the Columbia] is significantly outside of test database." How far outside? The final bullet will tell us.

This fourth-level bullet concluding the slide reports that the debris hitting the Columbia is estimated to be 1920/3 = 640 times larger than data used in the tests of the model! The correct headline should be "Review of Test Data Indicates Irrelevance of Two Models." This is a powerful conclusion, indicating that pre-launch safety standards no longer hold. The original optimistic headline has been eviscerated by the lower-level bullets.

Note how close readings can help consumers of presentations evaluate the presenter's reasoning and credibility.

The Very-Big-Bullet phrase fragment does not seem to make sense. No other VBB's appear in the rest of the slide, so this VBB is not necessary.

Spray On Foam Insulation, a fragment of which caused the hole in the wing

A model to estimate damage to the tiles protecting flat surfaces of the wing

Review of Test Data Indicates Conservatism for Tile Penetration

- **The existing SOFI on tile test data used to create Crater was reviewed along with STS-87 Southwest Research data**
  - **Crater overpredicted penetration of tile coating significantly**
    - **Initial penetration to described by normal velocity**
      - Varies with volume/mass of projectile (e.g., 200ft/sec for 3cu. In)
    - **Significant energy is required for the softer SOFI particle to penetrate the relatively hard tile coating**
      - Test results do show that it is possible at sufficient mass and velocity
    - **Conversely, once tile is penetrated SOFI can cause significant damage**
      - Minor variations in total energy (above penetration level) can cause significant tile damage
  - **Flight condition is significantly outside of test database**
    - **Volume of ramp is 1920cu in vs 3 cu in for test**

BOEING

Here "ramp" refers to foam debris (from the bipod ramp) that hit Columbia. Instead of the cryptic "Volume of ramp," say "estimated volume of foam debris that hit the wing." Such clarifying phrases, which may help upper level executives understand what is going on, are too long to fit on low-resolution bullet outline formats. PP demands the shorthand of acronyms, phrase fragments, and clipped jargon in order to get at least some information into the tight format.

Edward Tufte

Our models are irrelevant

Debris hitting the wing was 640x larger than the experimental data used to build these models

We have no clue what will happen on re-entry

Communication is key

Identify audience & setting

Identify key insight, main points of evidence, and assumptions

Organize into a story focussed on ⬆️

Create supporting visualizations

Revise to be as precise and concise as possible

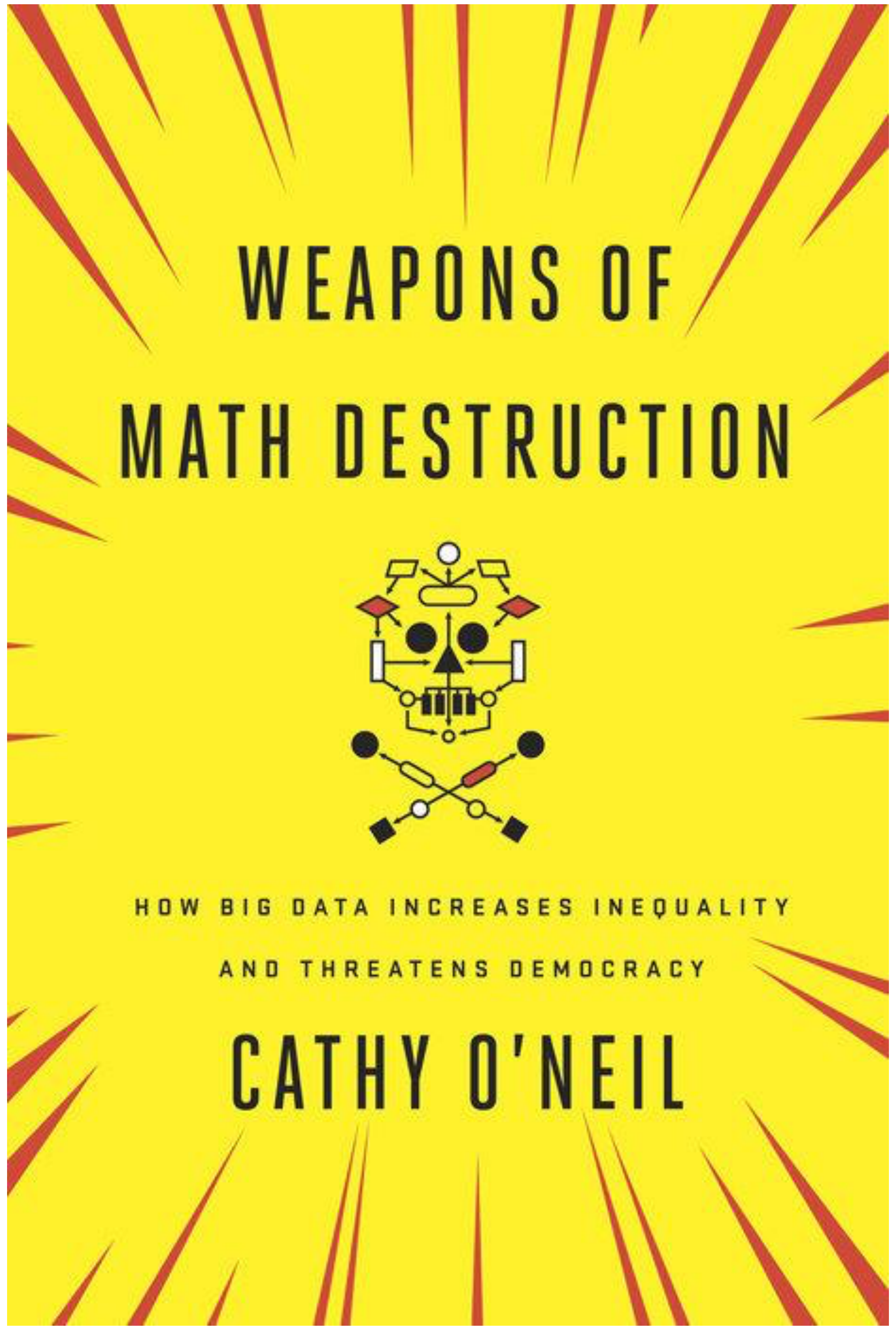
Ethical errors

Problems arising from data science and algorithms

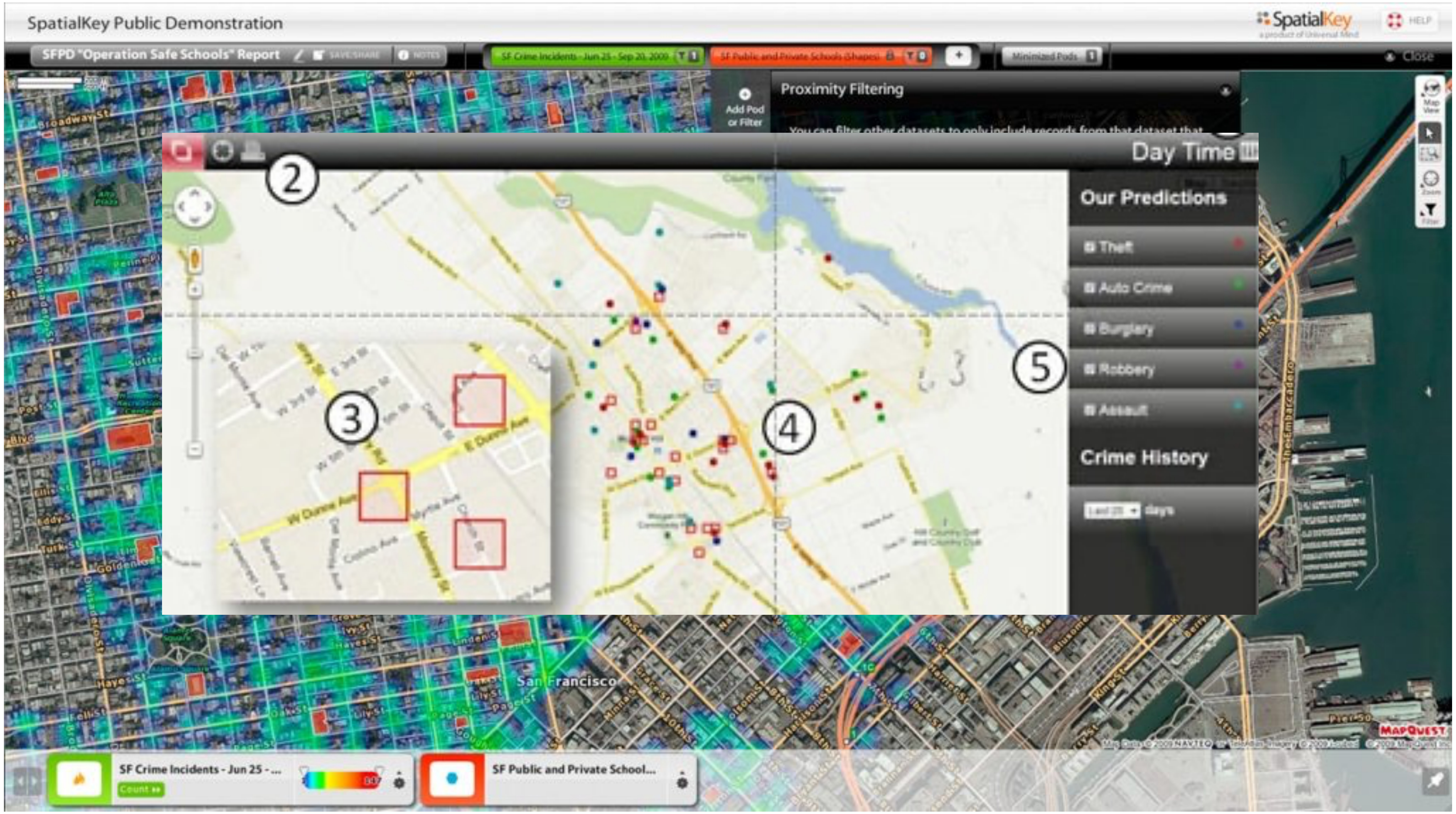
- Bias
- Ethics of use
- Accountability
- Societal impacts
- Economic impacts
- Environmental impacts

Don’t be a tool for creating WMDs

- Algorithms (and DS!) implement our biases, yet look objective
- Can implement our biases at scale
- Can have huge impacts on people’s lives
- Are not transparent or accountable to the people being impacted



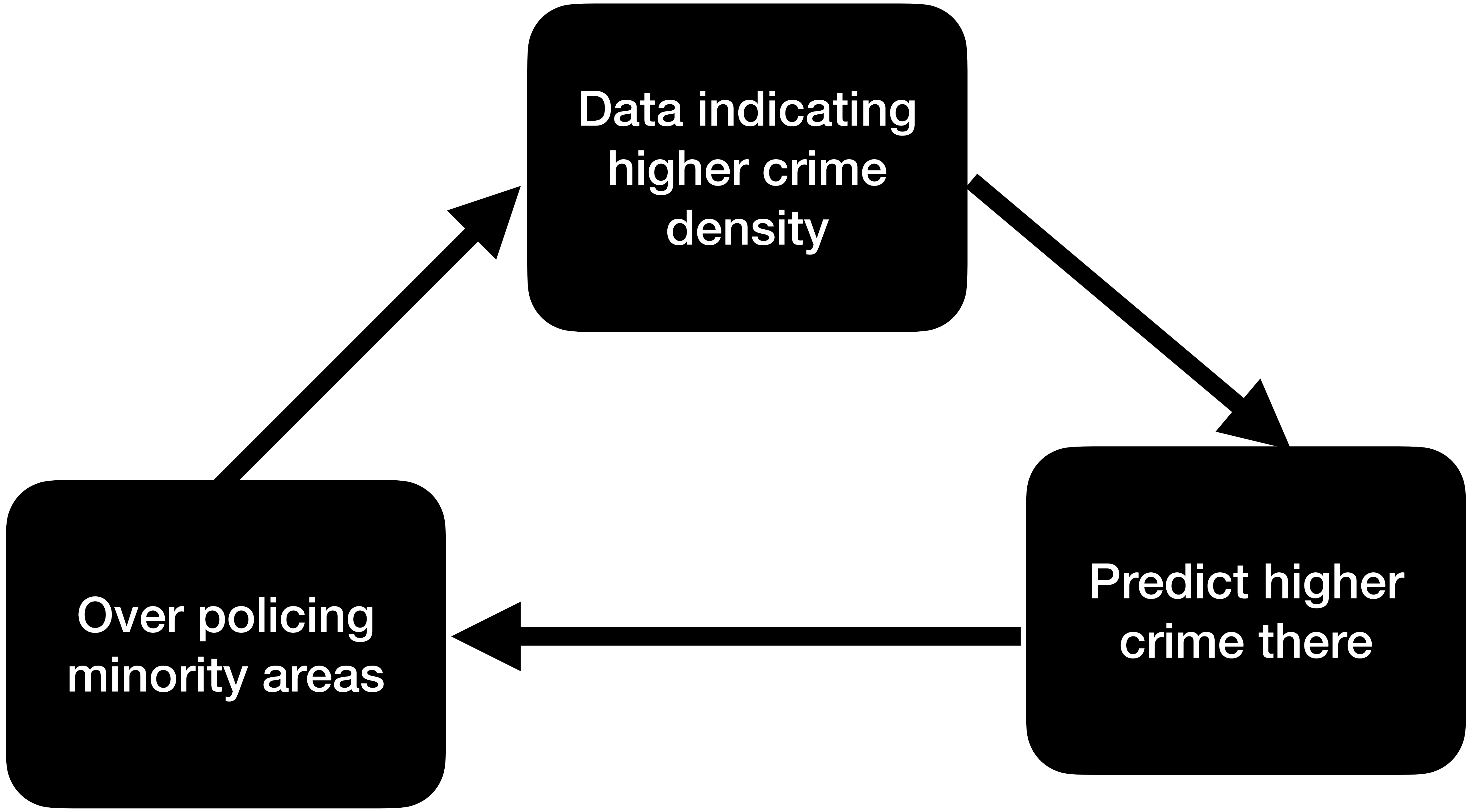
Predictive policing



https://americanpoliceofficersalliance.com/police-tech-predictive-technology/#prettyPhoto

Predictive policing & sentencing

Blacks arrested for possession at 4x the rate of whites
Usage rates the same



https://norml.org/marijuana/fact-sheets/racial-disparity-in-marijuana-arrests/ https://rss.onlinelibrary.wiley.com/doi/full/10.1111/j.1740-9713.2016.00960.x

https://themarkup.org/prediction-bias/2021/12/02/crime-prediction-software-promised-to-be-free-of-biases-new-data-shows-it-perpetuates-them

“A lot of times, people are talking about bias in the sense of equalizing performance across groups. They’re not thinking about the underlying foundation, whether a task should exist in the first place, who creates it, who will deploy it on which population, who owns the data, and how is it used?”

-Timnit Gebru

Problems with AI

Same same

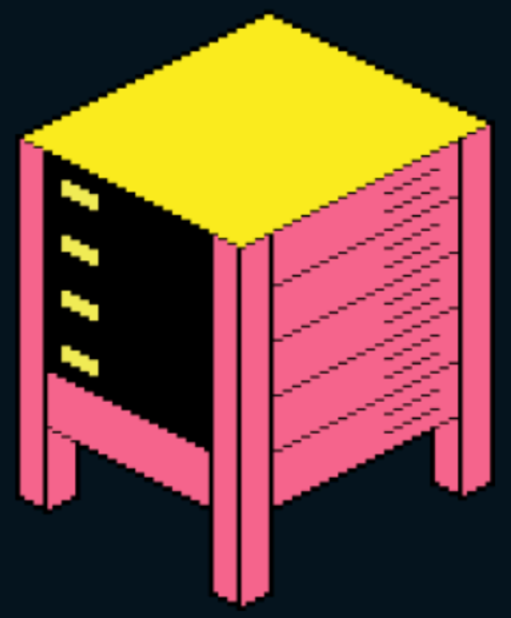
- Bias
- Ethics of use
- Accountability
- Societal impacts
- Economic impacts
- Environmental impacts

https://forms.gle/yN2j4u9D9bokXqMn8

four years. Google expects to spend $75 billion on AI infrastructure alone in 2025.

This isn't simply the norm of a digital world. It's unique to AI, and a marked departure from Big Tech's electricity appetite in the recent past. From 2005 to 2017, the amount of electricity going to data centers remained quite flat thanks to increases in efficiency, despite the construction of armies of new data centers to serve the rise of cloud-based online services, from Facebook to Netflix. In 2017, AI began to change everything. Data centers started getting built with energy-intensive hardware designed for AI, which led them to double their electricity consumption by 2023. The latest reports show that 4.4% of all the energy in the US now goes toward data centers.

The carbon intensity of electricity used by data centers was 48% higher than the US average.



Given the direction AI is headed—more personalized, able to reason and solve complex problems on our behalf, and everywhere we look—it's likely that our AI footprint today is the smallest it will ever be. According to new projections published by Lawrence Berkeley National Laboratory in December, by 2028 more than half of the electricity going to data centers will be used for AI. At that point, AI alone could consume as much electricity annually as 22% of all US households.

https://www.technologyreview.com/supertopic/ai-energy-package/

https://slate.com/technology/2024/10/uber-lyft-gig-workers-artificial-intelligence-wage-discrimination-jobs.html

SLATE News & Politics Culture Technology Business Life Advice Podcas

YES... HA HA HA... YES!

THE INDUSTRY

Why You Might Soon Be Paid Like an Uber Driver—Even If You're Not One

As gig workers' pay gets slashed by algorithms, experts warn that A.I.-driven wage systems mean that no one's paycheck is safe.

BY MOLLY GLICK OCT 13, 2024 • 5:45 AM

The San Francisco Standard

Latest News Politics Business Opinion Culture Food & Drink Sports Podcasts

Business

It's official. AI is already killing entry-level jobs

A Stanford study finds AI is slashing jobs for 22- to 25-year-olds, with software engineers hit hardest. New grads face a tougher market than ever.

Published Aug. 27, 2025•10:15am

Share

The AI jobpocalypse has begun.

For professions at risk of being replaced by AI, such as software engineering and customer service, entry-level employment has fallen 13% since 2022, a Stanford study (opens in new tab) has found.

The research, which analyzed millions of jobs using data from payroll processing firm ADP, painted an especially grim picture for tech. Employment for software developers ages 22 to 25 has fallen by almost 20% since ChatGPT's public release.

To hear it from Anthropic CEO Dario Amodei, this is merely the start. AI could wipe out half of entry-level white-collar jobs over the next five years, he told Axios (opens in new tab).

And then walk through this:

https://www.anthropic.com/research/labor-market-impacts

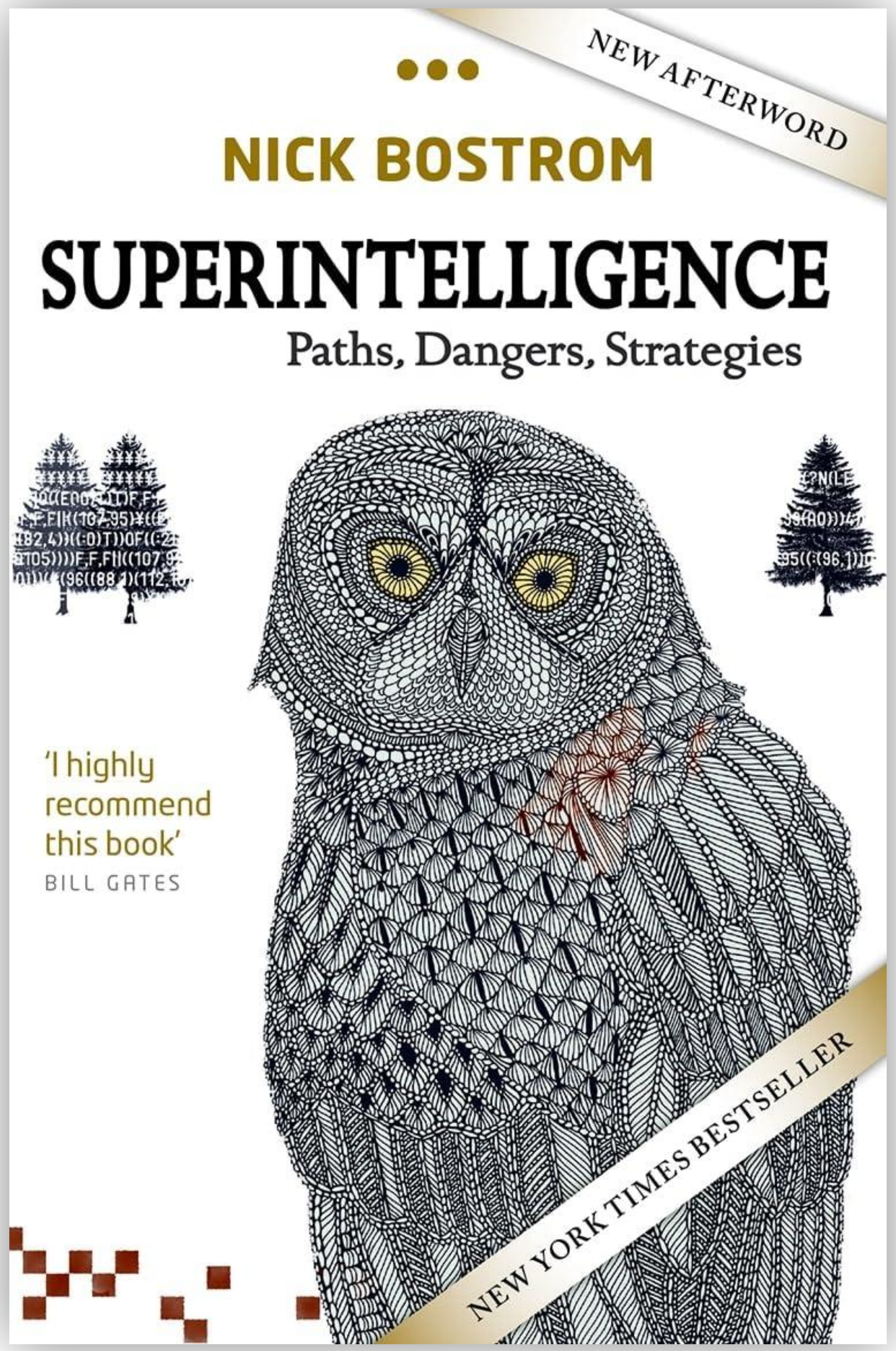
Who owns the stuff?

- AI is a powerful engine of economic inequality
- Copyright cases galore w.r.t training data… fair use or theft?
- Sell a monthly subscription everyone needs to keep their jobs against stiff competition
- This service actually creates that competition
- Instead of lots of jobs, the money (and training data!) goes to AI companies

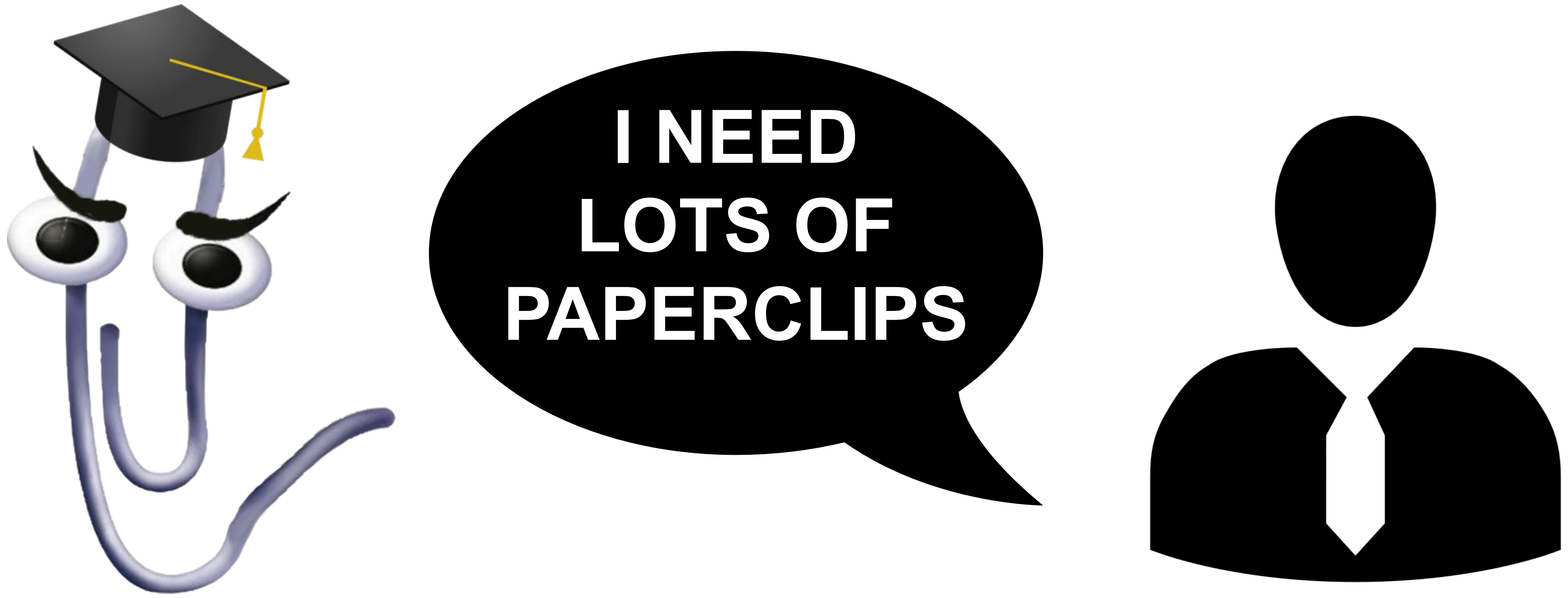
Problems with AI

New new

- Interpretability
- Alignment



Paperclip AI Thought Experiment



Orthogonality

Greater intelligence does not imply greater alignment to human values

Instrumental Convergence

All final goals pursued by sufficiently intelligent agents have subgoals in common

(Super)intelligent AI may (un)intentionally make decisions misaligned with human values even while pursuing harmless goals

Cognitive Superpowers

What Bostrom claims an AI would need to be able to do for world domination, human extinction, and other fun activities. What would make an AI a threat?

Task	Skillset
Hacking	Access and manipulate computational resources remotely
Technology Research	Develop technology with real-world possibilities
Economic Productivity	Perform economically productive activity to acquire resources
Strategizing	Achieve distant goals and overcome opposition via planning
Social Manipulation	Leverage human and institutional support via persuasion
Intelligence Amplification	Bootstrap its own capabilities independently

The alignment problem is fighting the law of unintended consequences

LLMs are trained to be polite and not upset users

Instagram's AI Chatbots Lie About Being Licensed Therapists (404 media)

Most AI Chatbots Easily Tricked Into Giving Dangerous Responses, Study Finds (the guardian)

LLMs make it easy to do some kinds of homework

Ghost Students Are Creating an 'Agonizing' Problem For California Colleges (sfgate.com)

Wonder what might go wrong here? Wonder why he wants this?

Zuckerberg's Grand Vision: Most of Your Friends Will Be AI (msn.com)

Broader questions

How can we be more right and less wrong?

Being aware of our biases; external review and checks to minimize

Thinking about tools; know their strengths and limitations

Worry about process and rigorous methods more than outcomes

Social factors influence decision making; good comm and procedures are essential

Be a systems level thinker! Stay curious! Question yourself and your team as much as possible